Assignment 1

Lumen Beauty and BakeBox

Findings, and the decisions submitted for approval

I reviewed the September livestreams and the August orders, and I am sending this note for approval. Each section states the question, the figure I computed, what it means, and the decision I recommend. Twelve sessions are included, with the repeated S07 row counted once. BakeBox is 2,800 orders, split between the live show and the storefront. GMV is what was sold. It is not the cash left after product cost.

1. Ranking a good stream

Is the session with the highest sales per viewer also the one with the highest peak audience, and which ranking should define a good stream?

Peak audience and sales per viewer do not point to the same session. S05 is first on the rate, at 18.08 CNY, but the room peaked at 379 people and the stream sold 6,852.76 CNY. S12 is almost the same rate, 18.03 CNY, with 1,427 people at the peak and 25,728.68 CNY sold. S11 is the result I would not repeat: 9,500 comments, and the lowest sales per viewer in the month, 11.88 CNY.

	S05	S12
GMV per viewer	18.08 CNY	18.03 CNY
People at the peak	379	1,427
Sold	6,852.76 CNY	25,728.68 CNY

I recommend we judge a good stream by the sales rate only when the audience is large, and that we repeat S12 rather than S05.

2. Orders for month two

Which two products grew fastest from week 1 to week 4, and what should we do with each in the second month?

Cloud Cream is both the fastest product and the largest one. It moved from 15,418 CNY in week 1 to 30,297 CNY in week 4, up 96.5 percent, and the gain held every week. Week 4 was about 152 units and 42 percent of that week's sales. Mist Toner also rose every week, from 2,467 CNY to 4,544 CNY, up 84.2 percent, but week 4 was only about 51 units. At 89 CNY it is a step before the cream, not a second lead product.

	Cloud Cream 50g	Mist Toner 150ml
Week 1 to week 4	15,418 to 30,297 CNY	2,467 to 4,544 CNY
Growth	Up 96.5%, every week	Up 84.2%, every week
Units in week 4	About 152	About 51
Note	42% of that week's sales	89 CNY, before the cream
Order I recommend	**About 150 units a week**	**About 50 units a week**

I recommend we order about 150 units of Cloud Cream a week, about 600 units over four weeks, and open every livestream with it. For Mist Toner I recommend one mention per stream and about 50 units a week, reviewed if sales do not stay near 4,544 CNY.

3. Whether the streams covered their cost

Did the September streams cover their cost, and which assumption would change that conclusion?

The 12 sessions ran 23 hours. At 800 CNY an hour for the host, the room, and samples, the bill is 18,400 CNY. The streams sold 182,907 CNY, 9.9 times that bill. The difference is not profit, because the hourly rate does not pay for the product. The month covers its streaming cost only if we keep more than 10.06 percent of stream sales.

Streaming bill	What the streams sold	Break-even margin
18,400 CNY	**182,907 CNY**	**10.06%**

A 30 percent margin keeps 54,872 CNY and covers the bill. An 8 percent margin keeps 14,633 CNY and does not. The conclusion depends on the margin, not on a small change in the hourly rate. At 1,600 CNY an hour the bill would be 36,800 CNY. A 30 percent margin would still cover it. An 8 percent margin would still fail at 800.

If the margin is	You keep	Against 18,400	Covered?
30% of stream GMV	54,872 CNY	Covers the bill	Yes
8% of stream GMV	14,633 CNY	Short of the bill	No

I recommend we approve the September streams as having covered their cost only if gross margin is above about 10 percent.

4. The apron, the bowl, and the spatula

Which three associations are the strongest, and what should we do with one of them?

On all 2,800 orders, keeping sets that appear in at least 5 percent of baskets, the three strongest associations are the same three products. In about 221 orders, a basket that already holds the mixing bowl and the spatula also holds the apron 99.1 percent of the time. That is about ten times what chance would produce. The bold row is the association I recommend we act on.

Rule	Support	Confidence	Lift
apron -> bowl + spatula	0.079	0.804	10.091
bowl + spatula -> apron	**0.079**	**0.991**	**10.091**
bowl -> apron + spatula	0.079	0.759	9.622

I recommend we sell the apron, the mixing bowl, and the spatula set as one bundle, on one product page and in the live script. Where the cart already has two of the three, we should offer the third.

5. Paper doilies

Paper doilies are the most bought product. Do they form a strong pair with anything else, and if not, what should a keep-or-drop decision rest on?

Paper doilies are the most purchased item, in 784 of 2,800 orders. They do not form a strong pair with any other product. Lift runs from 0.804 with sugar to 1.250 with cupcake cases, close to chance. The largest pair is 133 orders, short of the 140 required at the 5 percent line, so no doily rule qualifies. A bundle built from doilies would mostly name products that sell on their own.

Share of orders	Lift with other items	Rules at the 5% line
28%	**0.80 to 1.25**	**None**

I do not recommend a doily bundle. I recommend we keep or drop the item on its own profit: units, margin per unit, and the cost to store and pack. If the item itself is profitable, keep it as a low-priced extra. If it is not, delist it, despite a 28 percent share of orders.

6. Live show and storefront

Is there a rule that is strong on the live show and absent from the storefront, and should we use one combined list or one list per channel?

The same 5 percent line does not produce the same pairs in the two channels. On the live show, 816 orders, every order with a spatula also contains the apron: support 0.271, confidence 1.000, lift 3.692. That pair is absent from the storefront. The storefront, 1,984 orders, has its own pairs. A combined list still shows the apron set with a lift near 10, because the live orders are mixed in. Using that list on the website would carry a habit from the show into the storefront.

Live show, 816 orders

spatula -> apron

Support 0.271

Confidence 1.000

Lift 3.692

Present in every spatula order.

Storefront, 1,984 orders

piping bag -> nozzle set

Support 0.109, lift 9.185

whisk -> matcha powder

Support 0.100, lift 8.371

The apron pair does not appear.

I recommend two lists. On the live show, keep offering the spatula, the apron, and the mixing bowl together. On the storefront, use the baking pairs, and leave the apron kit off the main slot of the bundle page.

These six decisions are what I am asking you to approve for next month's buying, the live script, and the storefront page.

AI-use appendix

What did I compute myself, and where did I use help to draft the findings?

I did half of this assignment myself. Grok helped me draft the findings, and the parts of the write-up I had not finished. The numbers are the ones the notebook prints.

On S09, the file shows an average watch time of 252 on a stream that lasted 150 minutes. I left that figure as it was exported. Tasks 1.1 to 1.3 use sales, viewers, stream length, and weekly product sales. They do not use watch time, so the decisions above are the same whether that figure is corrected or left as it is.